

Raghav Khandelwal

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Education

Washington State University

Aug 2024 – Dec 2025

Master of Science in Computer Science

Pullman, WA

- **Relevant Coursework:** Distributed Systems, Advanced Algorithms, Database Systems, Machine Learning

Experience

Community Dreams Foundation

Mar 2026 – Present

Software Developer

Remote, USA

- Architect a TypeScript/Express REST API serving volunteer, staff, and admin permission tiers, eliminating 10+ hours/week of manual donor-record exports by building self-serve, audit-logged data access with JWT-based RBAC enforced through layered middleware guards
- Automate recurring payment processing and email campaign triggers across client-facing workflows by integrating Stripe and Mailchimp APIs with idempotent webhook handling into the donor management backend
- Cut client-facing wait times by 70% by building an asynchronous background job system with Redis and Bull message queues for report generation and email dispatch, decoupling long-running tasks from the request cycle
- Reduce production regressions by establishing automated testing and CI/CD with Jest, Supertest, and GitHub Actions, gating merges on passing unit and integration tests

DXC Technology

Jun 2022 – Nov 2022

Associate Professional Software Engineer

India

- Eliminated recurring integration failures between legacy and modern database environments by engineering a Python and SQL ETL pipeline extracting, transforming, and loading 6+ GB/month from DB2 into PostgreSQL
- Developed RESTful APIs for insurance policy operations sustaining 99.5% uptime at 500+ requests/min, publishing documented endpoint contracts consumed by downstream teams
- Cut environment setup time by 60% and standardized deployments across dev, staging, and production by containerizing 4+ enterprise applications with Docker on AWS (EC2, RDS, S3)
- Reduced query response latency from 4.2s to 800ms (81%) by optimizing SQL Server and DB2 performance through index tuning and join refactoring

Projects

Greenlight – Go REST API | Go, PostgreSQL, net/http

[GitHub](#)

- Prevented lost updates under concurrent writes by engineering a production-grade Go REST API on PostgreSQL with optimistic concurrency control via version-based conflict detection
- Protected endpoints from abuse and credential misuse by implementing stateful token authentication and per-client rate limiting middleware, with graceful shutdown and structured JSON logging for operational reliability
- Enabled efficient filtered, sorted, and paginated queries across large result sets by building full-text search over GIN-indexed PostgreSQL columns

Snowpack Prediction – AgAid Hackathon | Python, PyTorch, Flask, React

[Link](#)

- Achieved 87.8% accuracy predicting Snow Water Equivalent across 9M+ spatio-temporal records from 8 meteorological datasets by designing a Transformer-based neural network in PyTorch
- Delivered real-time predictions to end users by deploying a Flask and React service with live CSV ingestion and interactive SWE visualizations

AI Agent | TypeScript, Node.js, Python, OpenAI API, RAG, PyTest

[GitHub](#)

- Engineered a multi-tool AI agent with a RAG pipeline, vector embeddings, and OpenAI API integration across TypeScript and Python services, validated by a PyTest suite covering retrieval and tool-routing paths
- Improved retrieval relevance by 30% on a 100-query evaluation set by implementing semantic retrieval and reusable prompt configurations across content workflows

Technical Skills

Languages: Go, Python, TypeScript, JavaScript, SQL

Backend: Node.js, Express.js, Flask, REST APIs, WebSockets, JWT, RBAC, Rate Limiting, Optimistic Concurrency Control, Full-Text Search, Message Queues (Bull)

Testing & CI/CD: PyTest, Jest, Supertest, Unit & Integration Testing, GitHub Actions

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Git, Linux

Databases & Caching: PostgreSQL, Redis, DB2

AI/ML: PyTorch, OpenAI API, RAG, Vector Embeddings

Frontend: React, Next.js